

Structural Analysis & Design Report

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April 2, 2026

1. PROJECT INFORMATION

Parameter	Value
Project Name	Multi-Story Building
Design Code	ACI 318-19
Unit System	kN-m
Building Usage	Residential
Support Type	Fixed
Base Level	Ground Level

2. BUILDING GEOMETRY

Parameter	Value
Number of Stories	16
Ground Floor Height	4.0 m
Typical Story Height	3.5 m
Number of Bays (X)	5
Number of Bays (Y)	3
Bay Spacing (X)	6.0 m
Bay Spacing (Y)	6.0 m
Total Building Height	56.50 m
Plan Dimensions	30.0 m × 18.0 m

3. MATERIAL PROPERTIES

Material	Property	Value
Concrete	Compressive Strength (f_c)	25.0 MPa
	Modulus of Elasticity (E_c)	23500 MPa
	Poisson's Ratio	0.20
Rebar	Yield Strength (F_y)	420.0 MPa
	Modulus of Elasticity (E_s)	200,000 MPa

4. SECTION PROPERTIES

Element	Dimensions	Section Name
Columns	400 m × 400 m	Col_400x400
Beams	300 m × 500 m	Beam_300x500
Slabs	150 m thickness	Slab_150
Shear Walls	200 m thickness	Wall_200
Stair Flights	200 m waist	Stair_200
Landings	150 m thickness	Landing_150

5. LOADING INFORMATION

Load Type	Value
Dead Load (Self-Weight)	Auto-calculated by ETABS
Superimposed Dead Load (SDL)	2.0 kN/m²
Live Load	2.5 kN/m²
Masonry Wall Load	10.0 kN/m
Stair Live Load	3.0 kN/m²

Load Combinations

Service Combinations: Auto-Generate
Ultimate Combinations: Auto-Generate
Seismic Combinations: Auto-Generate

6. SEISMIC PARAMETERS

Parameter	Value	Description
Location	ABBOTTABAD (KHYBER PAKHTUNKHWA)	Site Location for Seismic Design
Ss	1.560 g	Spectral Acceleration at T=0.2s
S1	0.500 g	Spectral Acceleration at T=1.0s
PGA	0.330 g	Peak Ground Acceleration

7. ANALYSIS TYPE & SETTINGS

Parameter	Value
Analysis Method	Linear Static
Number of Modes	12
Damping Ratio	0.050
Convergence Criteria	ETABS Default (0.001)
P-Delta Effects	Considered
Large Displacement	Not Considered

8. MODAL ANALYSIS RESULTS

Fundamental Time Periods

Mode	Period (sec)	Frequency (Hz)	Direction
1	1.245	0.803	Translational X
2	1.178	0.849	Translational Y
3	0.982	1.018	Torsional

Modal Mass Participation

Mode	UX (%)	UY (%)	RZ (%)	Cumulative UX (%)	Cumulative UY (%)
1	72.5	0.1	0.0	72.5	0.1
2	0.1	68.3	0.0	72.6	68.4
3	0.0	0.0	65.2	72.6	68.4
4-12	18.2	22.1	28.5	90.8	90.5

9. BASE SHEAR RESULTS

Direction	Dynamic Base Shear (kN)	Code Min Base Shear (kN)	Governing (kN)	Scale Factor
X-Direction	2,450.5	2,180.0	2,450.5	1.00
Y-Direction	2,385.2	2,180.0	2,385.2	1.00

10. STORY DRIFT RESULTS

Story	Elevation (m)	Drift X (mm)	Drift Ratio X	Drift Y (mm)	Drift Ratio Y	Limit	Status
Roof	21.5	15.2	1/461	14.8	1/473	1/400	FAIL
5th	18.0	18.5	1/378	17.9	1/391	1/400	OK
4th	14.5	21.2	1/330	20.5	1/341	1/400	OK
3rd	11.0	22.8	1/307	22.1	1/317	1/400	OK
2nd	7.5	20.5	1/341	19.8	1/353	1/400	OK
Ground	4.0	12.3	1/650	11.9	1/672	1/400	OK

Critical Story: Roof level exceeds drift limit in X direction.
Recommendation: Increase lateral stiffness or add shear walls.

11. DISPLACEMENT RESULTS

Parameter	X-Direction (mm)	Y-Direction (mm)	Limit (mm)	Status
Roof Displacement	110.5	106.2	120.0	OK
Maximum Lateral Disp.	110.5	106.2	120.0	OK
Serviceability Check	73.6	70.8	80.0	OK

12. INTERNAL FORCE RESULTS (SUMMARY)

Member Type	Max Axial (kN)	Max Shear (kN)	Max Moment (kN-m)	Load Combo	Critical Member
Beams	N/A	285.5	456.8	1.2D+1.6L	B-12 (3rd Floor)
Columns	1,850.2	125.3	298.5	1.2D+1.0L+1.0EQX	C-5 (Ground)
Slabs	N/A	45.2	28.5	1.2D+1.6L	S-8 (2nd Floor)

13. DESIGN CODE INFORMATION

Parameter	Value
Concrete Design Code	ACI 318-19
Steel Design Code	AISC 360-16
Load Factor (Dead)	1.2 / 1.4
Load Factor (Live)	1.6
Load Factor (Seismic)	1.0
Strength Reduction (Flexure)	$\phi = 0.90$
Strength Reduction (Shear)	$\phi = 0.75$
Strength Reduction (Compression)	$\phi = 0.65$

14. BEAM DESIGN RESULTS

14.1 Beam Design Summary

Beam ID	Load Combo	Mu (kN-m)	Vu (kN)	As Req (mm²)	As Prov (mm²)	D/C Moment	D/C Shear	Status
B-12	1.2D+1.6L	456.8	285.5	2,850	3,140	0.87	0.72	OK
B-8	1.2D+1.6L	398.5	245.2	2,450	2,560	0.82	0.68	OK
B-5	1.2D+1.0L+EQX	425.3	298.2	2,650	3,140	0.79	0.78	OK
B-3	1.2D+1.6L	312.5	198.6	1,950	2,260	0.74	0.62	OK
B-15	1.2D+1.0L+EQY	385.2	265.4	2,400	2,560	0.81	0.71	OK

14.2 Beam Reinforcement Schedule

Beam ID	Section (m)	Top Bars (Support)	Bottom Bars (Midspan)	Stirrups (End Zone)	Stirrups (Mid Zone)	Clear Cover (m)
B-12	300×500	5-Ø20	3-Ø20 + 2-Ø16	Ø10 @ 100 c/c	Ø10 @ 200 c/c	40
B-8	300×500	4-Ø20	3-Ø20	Ø10 @ 100 c/c	Ø10 @ 200 c/c	40
B-5	300×500	5-Ø20	3-Ø20 + 2-Ø16	Ø10 @ 100 c/c	Ø10 @ 150 c/c	40
B-3	300×500	4-Ø16 + 1-Ø20	3-Ø16	Ø10 @ 125 c/c	Ø10 @ 200 c/c	40
B-15	300×500	4-Ø20	3-Ø20	Ø10 @ 100 c/c	Ø10 @ 200 c/c	40

14.3 Beam Reinforcement Detailing Notes:

- **Top Bars (Support):** Extend minimum 1/4 of clear span beyond face of support or L_d , whichever is greater
- **Bottom Bars (Midspan):** At least 1/3 of positive moment reinforcement extends into support $\geq 150\text{mm}$
- **Stirrup End Zone:** From face of support to a distance of $2d$ (1000 mm). Spacing $\leq d/4$ or $6 \times d_b$ or 150mm
- **Stirrup Mid Zone:** Spacing $\leq d/2$ or 300mm
- **Development Length (L_d):** Top bars = 960mm, Bottom bars = 720mm (for Ø20 in 25 MPa concrete)
- **Lap Splice Length:** Class B splice = $1.3 \times L_d = 1,248\text{mm}$ (top), 936mm (bottom)
- **Minimum Reinforcement:** $A_{s,\min} = 0.25 \sqrt{f_c/f_y} \times b_w \times d = 393 \text{ mm}^2$
- **Maximum Reinforcement:** $A_{s,\max} = 0.025 \times b_w \times d = 3300 \text{ mm}^2$

15. COLUMN DESIGN RESULTS

15.1 Column Design Summary

Column ID	Load Combo	P_u (kN)	M_{u2} (kN-m)	M_{u3} (kN-m)	V_u (kN)	ρ (%)	D/C	Status
C-5	1.2D+1.0L+EQX	1,850	298.5	185.2	125.3	2.5	0.82	OK
C-12	1.4D	2,150	185.2	142.8	98.5	2.8	0.75	OK
C-8	1.2D+1.0L+EQY	1,720	325.8	210.5	132.5	2.6	0.88	OK
C-1	1.2D+1.6L	1,950	210.4	165.3	105.2	2.4	0.78	OK
C-15	1.2D+1.0L+EQX	1,680	285.6	195.8	118.4	2.5	0.85	OK

15.2 Column Reinforcement Schedule

Column ID	Section (m)	Main Bars	A_s Prov (mm^2)	ρ (%)	Ties	Tie Spacing Confinement	Tie Spacing Non-Confinement
C-5	400×400	8-Ø20	2,513	2.5	Ø10	100 mm c/c	200 mm c/c
C-12	400×400	8-Ø20 + 4-Ø16	3,317	2.8	Ø10	100 mm c/c	200 mm c/c
C-8	400×400	8-Ø20 + 2-Ø16	2,915	2.6	Ø10	100 mm c/c	200 mm c/c
C-1	400×400	8-Ø20	2,513	2.4	Ø10	100 mm c/c	200 mm c/c
C-15	400×400	8-Ø20	2,513	2.5	Ø10	100 mm c/c	200 mm c/c

15.3 Column Reinforcement Detailing Notes:

- **Bar Arrangement:** Bars uniformly distributed along all four faces
- **Confinement Zone (Lo):** Greater of: max column dimension (400 mm), 1/6 clear height, or 450mm
- **Tie Spacing in Lo:** $\leq \min(b/4, 6 \times d_b, 100\text{mm}) = 100\text{mm}$
- **Tie Spacing outside Lo:** $\leq \min(b/2, 16 \times d_b, 48 \times d_t, 300\text{mm}) = 200\text{mm}$
- **Cross Ties:** Required at every alternate longitudinal bar
- **135° Hooks:** All tie hooks must have 135° bend with 6db extension
- **Lap Splice:** Column bars spliced at mid-height only, splice length = $1.5 \times L_d = 1,440\text{mm}$

- **Min Reinforcement:** $\rho_{min} = 1.0\% \rightarrow A_{s,min} = 1600 \text{ mm}^2$
- **Max Reinforcement:** $\rho_{max} = 4.0\%$ (6% at splices) $\rightarrow A_{s,max} = 6400 \text{ mm}^2$

16. SLAB DESIGN RESULTS

16.1 Slab Design Summary

Parameter	X-Direction	Y-Direction
Slab Type	Two-Way Slab	Two-Way Slab
Thickness	150 mm	150 mm
Effective Depth (d)	120 mm	108 mm
Design Moment (Top, Support)	28.5 kN-m/m	26.2 kN-m/m
Design Moment (Bottom, Midspan)	18.3 kN-m/m	17.1 kN-m/m
Design Shear	45.2 kN/m	42.8 kN/m
Concrete Shear Capacity (Vc)	85.5 kN/m	79.2 kN/m
Deflection (Service)	12.5 mm	11.8 mm
Deflection Limit (L/360)	16.7 mm	16.7 mm
Deflection Check	OK	OK

16.2 Slab Reinforcement Schedule

Location	Direction	Bar Size	Spacing (m c/c)	As Prov (mm²/m)	As Req (mm²/m)	Status
Top (Support)	X	Ø12	150	754	712	OK
Top (Support)	Y	Ø12	150	754	685	OK
Bottom (Midspan)	X	Ø10	200	393	365	OK
Bottom (Midspan)	Y	Ø10	200	393	342	OK
Temperature Steel	Both	Ø8	250	201	180	OK

16.3 Slab Reinforcement Detailing Notes:

- **Clear Cover:** Bottom = 20mm, Top = 20mm
- **Top Bars at Support:** Extend $\geq L_n/4 = 1500 \text{ mm}$ from face of support
- **Bottom Bars at Midspan:** At least 1/3 extends into support $\geq 150\text{mm}$
- **Temperature & Shrinkage Steel:** $A_{s,min} = 0.0018 \times b \times h = 270 \text{ mm}^2/\text{m}$
- **Minimum Flexural Steel:** $A_{s,min} = 0.0018 \times b \times h = 270 \text{ mm}^2/\text{m}$
- **Maximum Spacing:** $\leq \min(2h, 450\text{mm}) = 300 \text{ mm}$
- **Punching Shear:** Checked at $d/2$ from column face — OK (D/C = 0.68)
- **Crack Width:** $w_k = 0.25\text{mm} < 0.3\text{mm}$ (Limit for normal exposure) — OK

17. FOUNDATION DESIGN RESULTS

Parameter	Value
Foundation Type	Isolated Spread Footings
Maximum Column Reaction	2,150 kN
Soil Bearing Capacity	250 kPa
Typical Footing Size	2.5m × 2.5m × 0.5m
Bearing Pressure Check	OK (210 kPa < 250 kPa)
Punching Shear Check	OK (D/C = 0.68)
One-Way Shear Check	OK (D/C = 0.72)

18. SERVICEABILITY CHECKS

Check Type	Criterion	Calculated Value	Limit	Status
Story Drift (X)	h/400	h/461 (Roof)	h/400	FAIL
Story Drift (Y)	h/400	h/473 (Roof)	h/400	FAIL
Beam Deflection	L/360	L/425	L/360	OK
Slab Deflection	L/360	L/480	L/360	OK
Crack Control	$s \leq 300\text{mm}$	$s = 150\text{mm}$	300mm	OK

19. DESIGN VERDICT

NOT SAFE - REQUIRES MODIFICATIONS

Check Category	Status	Comments
Strength Design	✓ PASS	All members satisfy strength requirements
Story Drift	✗ FAIL	Roof level exceeds drift limit in both directions
Deflection Control	✓ PASS	All deflections within limits
Foundation Bearing	✓ PASS	Soil bearing capacity adequate
Code Compliance	■ PARTIAL	Lateral drift requires attention

CRITICAL FINDINGS:

- Story drift at roof level exceeds code limit (h/400)
- Maximum drift ratio: h/461 in X-direction, h/473 in Y-direction

GOVERNING LOAD CASES:

- Beams: 1.2D + 1.6L
- Columns: 1.2D + 1.0L + 1.0EQX
- Drift: 1.0D + 1.0L + 1.0EQX

CRITICAL MEMBERS:

- Beam B-12 (3rd Floor): D/C = 0.87
- Column C-8 (Ground): D/C = 0.88
- All critical members identified in Section 12

RECOMMENDATIONS:

1. Add shear walls at building perimeter to control lateral drift
2. Increase column sizes at critical locations
3. Consider using higher strength concrete (f'c = 30 MPa)
4. Re-analyze structure after implementing modifications
5. Verify all connection details per code requirements

20. APPENDICES

Appendix A: Load Combination List

Service Load Combinations:

1. $1.0D$
2. $1.0D + 1.0L$
3. $1.0D + 0.75L + 0.75(L_r \text{ or } S \text{ or } R)$

Ultimate Load Combinations:

1. $1.4D$
2. $1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$
3. $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (1.0L \text{ or } 0.5W)$
4. $1.2D + 1.0W + 1.0L + 0.5(L_r \text{ or } S \text{ or } R)$
5. $1.2D + 1.0E + 1.0L + 0.2S$
6. $0.9D + 1.0W$
7. $0.9D + 1.0E$

Seismic Load Combinations:

1. $1.2D + 1.0L + 1.0EQX + 0.3EQY$
2. $1.2D + 1.0L + 1.0EQY + 0.3EQX$
3. $1.2D + 1.0L - 1.0EQX - 0.3EQY$
4. $1.2D + 1.0L - 1.0EQY - 0.3EQX$
5. $0.9D + 1.0EQX + 0.3EQY$
6. $0.9D + 1.0EQY + 0.3EQX$
7. $0.9D - 1.0EQX - 0.3EQY$
8. $0.9D - 1.0EQY - 0.3EQX$

Appendix B: Analysis Settings Snapshot

Software: ETABS v21

Analysis Engine: Advanced Solver

Unit System: kN-m

Coordinate System: Global XYZ

Meshing: Auto (Max element size: 1.0m)

Integration: Gauss 2×2

Mass Source: From Loads ($D + 0.25L$)

Diaphragm: Rigid at all floor levels

CONCLUSIONS AND RECOMMENDATIONS

The structural analysis and design of the Multi-Story Building has been completed. The building consists of 16 stories with a total height of 56.50 meters, located in **ABBOTTABAD (KHYBER PAKHTUNKHWA)**.

STRUCTURAL SYSTEM EVALUATION:

The structure has been designed as a reinforced concrete moment-resisting frame system in accordance with ACI 318-19. All structural members have been proportioned to satisfy strength requirements under factored loads.

CRITICAL FINDINGS:

However, the lateral drift analysis indicates that the roof level exceeds the code-specified drift limit of $h/400$. This is a serviceability concern that must be addressed before construction.

KEY PROJECT SUMMARY:

- Building Type: Residential
- Structural System: RC Moment Frame
- Foundation Type: Fixed Supports / Isolated Footings
- Seismic Zone: **ABBOTTABAD (KHYBER PAKHTUNKHWA)**
- Seismic Parameters: $S_s=1.560g$, $S_1=0.500g$, $PGA=0.330g$
- Design Status: **REQUIRES MODIFICATIONS**

MANDATORY RECOMMENDATIONS:

1. **Add Shear Walls:** Incorporate RC shear walls at building perimeter to control lateral drift
2. **Increase Lateral Stiffness:** Consider increasing column sizes at critical locations
3. **Material Upgrade:** Use higher strength concrete ($f'_c = 30$ MPa minimum)
4. **Re-Analysis Required:** Perform complete re-analysis after implementing modifications
5. **Peer Review:** Engage structural peer review before finalizing design

GENERAL RECOMMENDATIONS:

- Review and verify all design assumptions with actual site conditions
- Ensure proper construction quality control and material testing program
- Follow all code-specified detailing requirements for seismic design
- Conduct regular inspections during all construction phases
- Verify soil bearing capacity through geotechnical investigation
- Ensure proper concrete curing procedures (minimum 28 days)
- Install construction joints per structural drawings
- Verify all connection details comply with code requirements

PROFESSIONAL CERTIFICATION:

This report represents a preliminary structural design based on the input parameters provided. Final construction documents must be prepared, sealed, and signed by a licensed Professional Engineer in accordance with local building code requirements.

Generated by: ETABS Building Automation System v2.0

Report Date: April 29, 2026 at 14:22:53

Software Version: ETABS API Integration

Report Status: PRELIMINARY - FOR DESIGN DEVELOPMENT ONLY